RIYADH

HeatReady

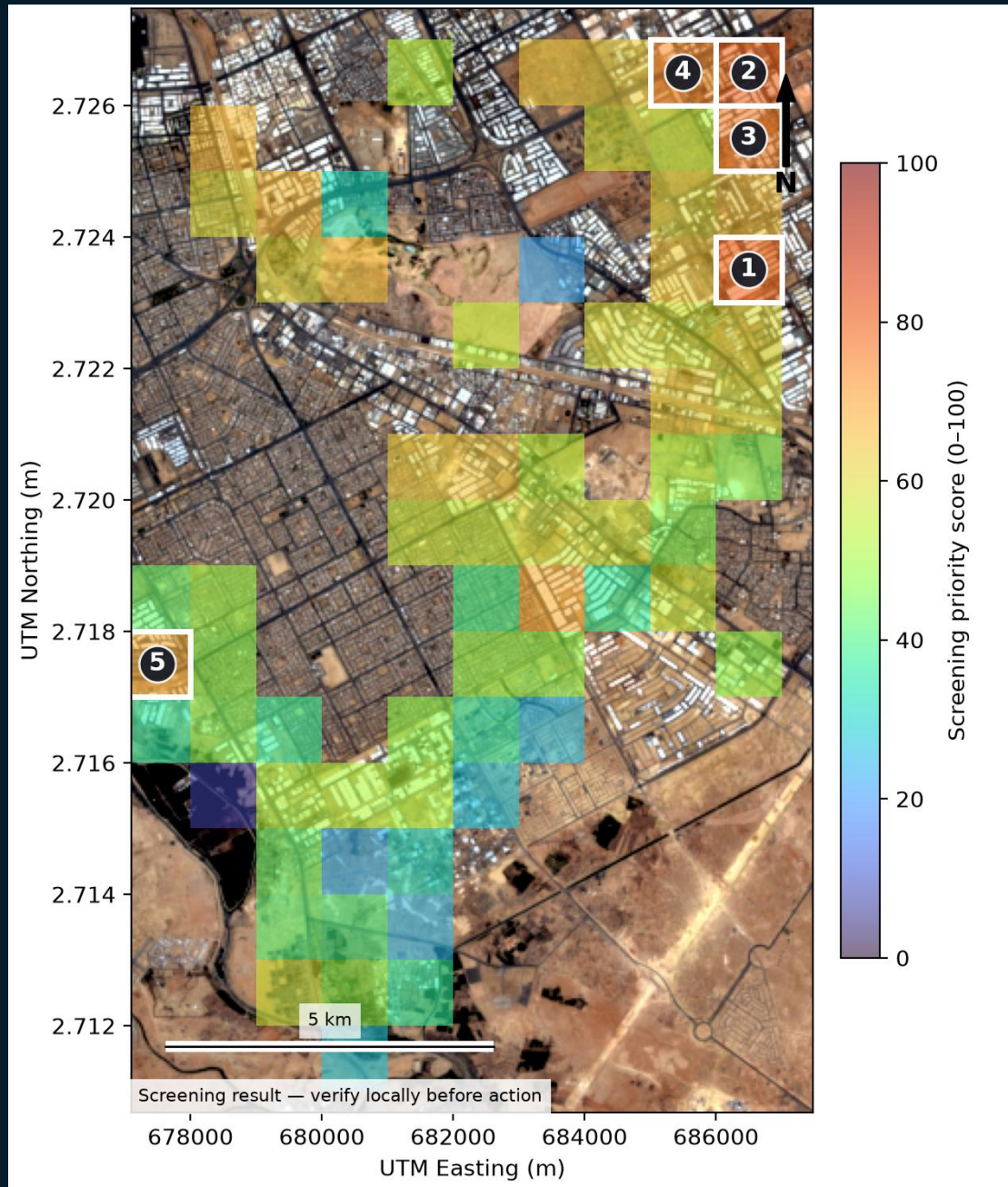
From satellite evidence to a clear municipal inspection priority

ONE DECISION

Which 1 km zones should a planning and heat-resilience team inspect first?

Milkaholics • Urban Expansion, Land Use Change & Heat Risk • United Arab Emirates

SATELLITE-DERIVED PRIORITY MAP



Two clear owners for one testable decision

YK

Yusuf Karanib

Project Lead and Data/Method Presenter

Project responsibility

- Coordinates the final submission and live demonstration.
- Explains the problem, EO data, method, score and live demonstration.

Project expertise: EO workflow, data processing and reproducibility.

RM

Raed Mardoum

Validation, Workflow and Impact Lead

Finals responsibility

- Explains validation evidence, limitations and safe interpretation.
- Explains innovation, expected impact and the municipal pilot plan.

Project responsibility: validation interpretation, decision workflow and pilot design.

The problem is not missing maps — it is a fragmented decision

168.30 km²
SOUTHEASTERN RIYADH STUDY AREA

1 km
TRANSPARENT SCREENING CELLS

1 user
MUNICIPAL PLANNING + HEAT TEAM

1 action
INSPECT LOCALLY BEFORE INTERVENTION

CURRENT PAIN

Growth, heat, population and vegetation sit in separate layers. A planner still needs a short, explainable list of places to inspect.

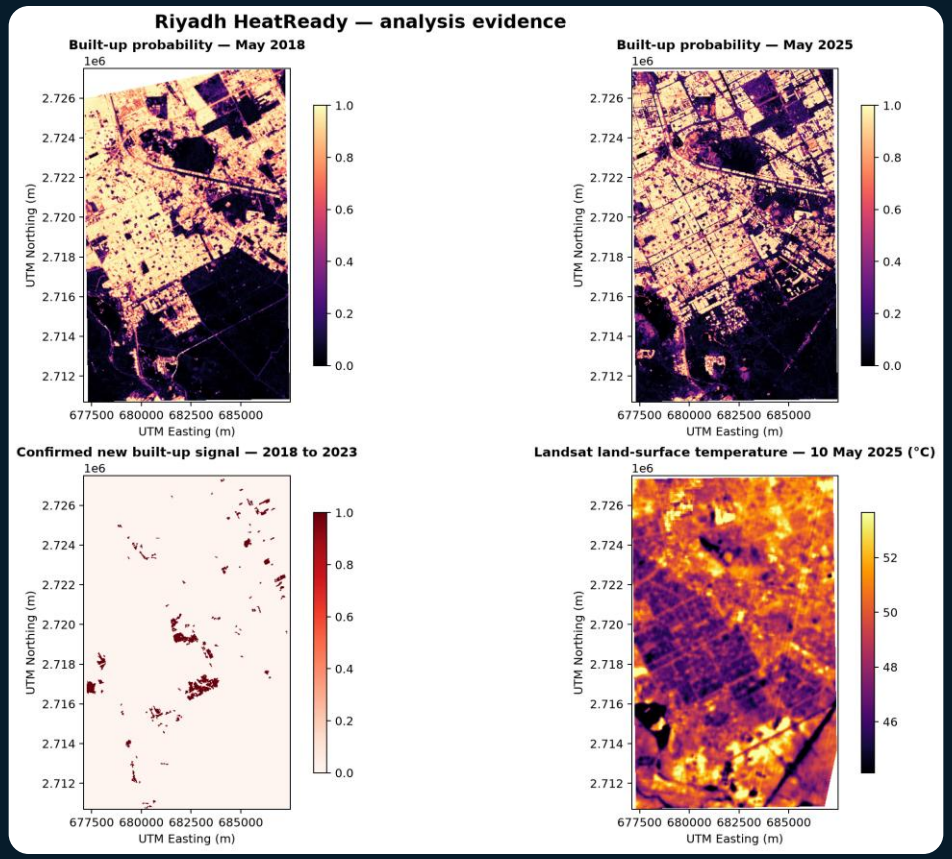
TESTABLE MVP

A ranked layer, dashboard and GIS files that show the score and its evidence — without automatically selecting a cooling intervention.

Honest gap: no municipal user interview has happened yet. The workflow and value must be tested during incubation.

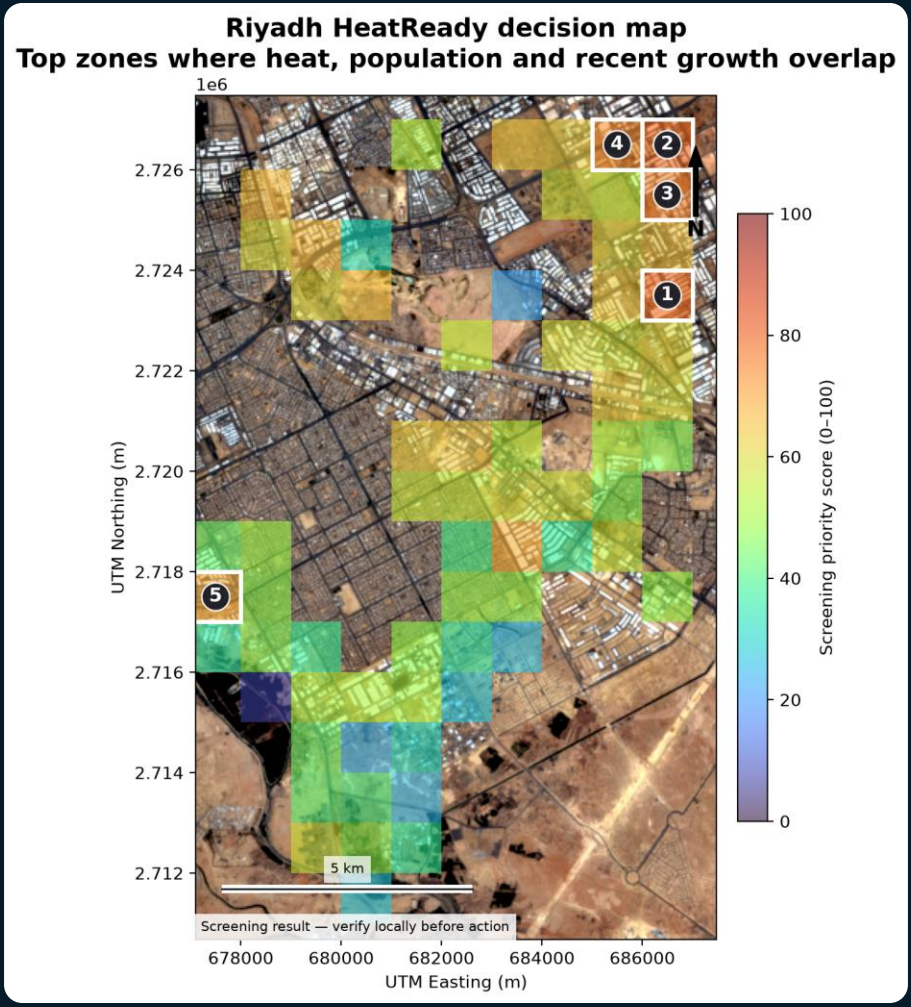
Six open sources, one conservative chain

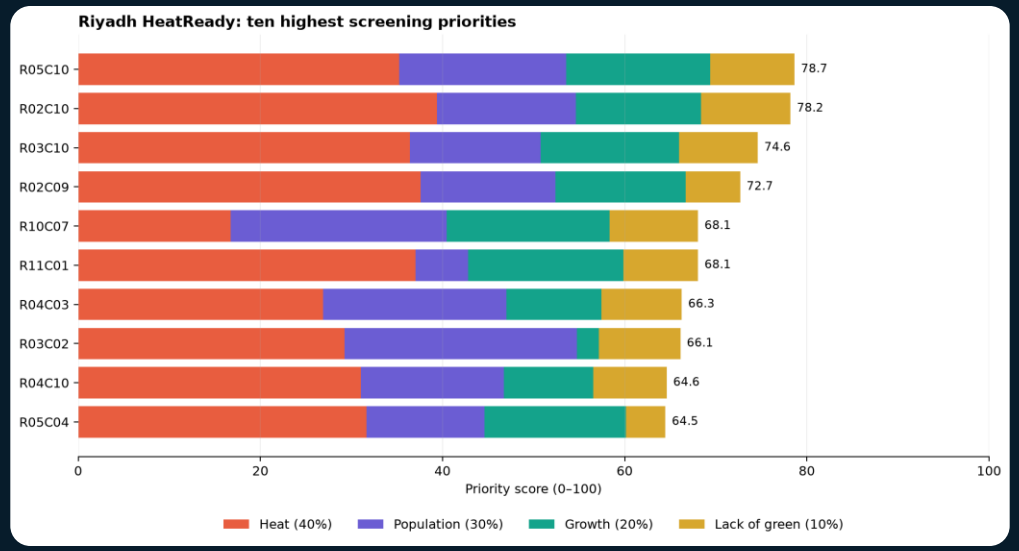
Sentinel-2	Optical reflectance, vegetation and built features
Sentinel-1	Radar structure to reduce bright-desert confusion
Annual LULC	Six maps propose 2018–2023 built-up change
Landsat 8	Quality-masked 2025 land-surface temperature
WorldPop	Modelled 2025 population exposure
Planet Tanager	426-band material context and failure-mode test



Annual change proposes growth → separate 2025 optical + radar evidence confirms it → heat, population and vegetation rank eligible cells.

A ranked map a planner can inspect and export





3.74 km²
CONFIRMED GROWTH SIGNAL

14 / 67
HIGH / ELIGIBLE CELLS

78.7
TOP SCORE: R05C10

Demo path: open dashboard → select top zone → inspect heat, population, growth and green components → export CSV/GeoJSON/GeoTIFF.

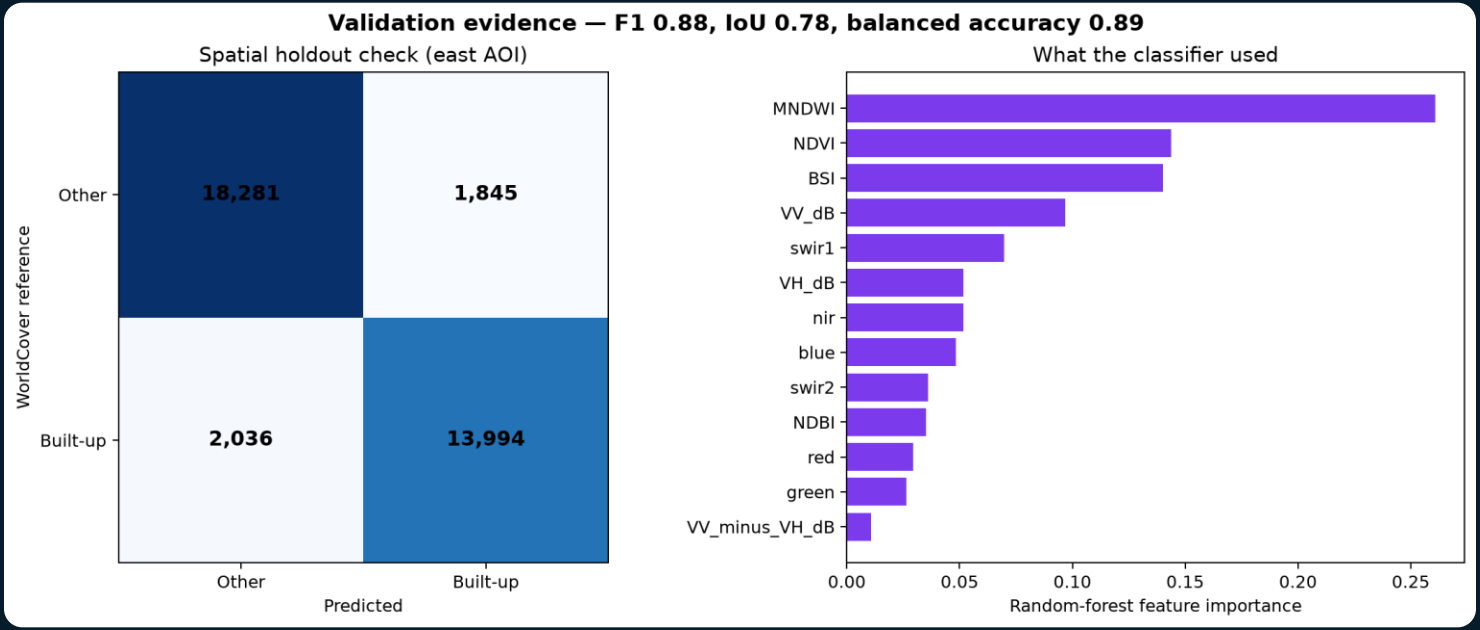
Measured agreement, explicit assumptions, bounded claims

0.88
F1 AGREEMENT

0.78
INTERSECTION OVER UNION

0.89
BALANCED ACCURACY

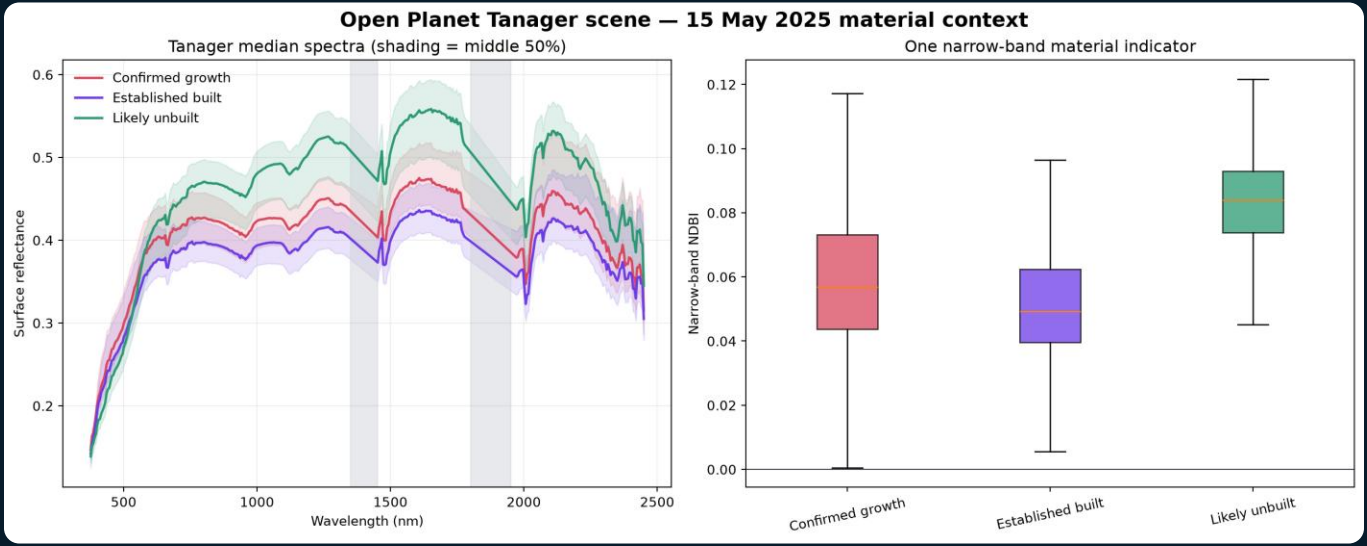
These compare the model with a spatially separate WorldCover reference. They are not field accuracy.



LIMITS JUDGES SHOULD HEAR

- Land-surface temperature is not air temperature or personal heat exposure.
- WorldPop is modelled; it is not a live census or worker count.
- Bright desert and roofs can still confuse classification.
- Score weights are PoC assumptions that a real user must review.

The value is the evidence chain — not a black-box score



WHAT IS DIFFERENT

- Annual maps propose change; independent optical and radar evidence confirms it.
- Tanager tests why a simple high-NDBI-means-built rule can fail in bright desert.
- Every ranked cell keeps visible score components and GIS-ready evidence.

Careful claim: one Tanager date adds material context. It does not prove change or identify specific materials.

POTENTIAL IMPACT

Reduce time spent comparing separate datasets and create a repeatable way to select field-inspection priorities. This is a hypothesis until real users test it.

Turn the PoC into a measured municipal pilot

1

Interview

Confirm the decision, current workflow and acceptable false alarms with at least three real users.

2

Validate

Inspect the top five zones, record useful results and false alarms, and compare with local plans.

3

Adapt

Agree on weights, add worker exposure and city-plan constraints, then rerun the ranking.

4

Scale

Test a repeatable update in another Riyadh area before any operational deployment claim.

THE FINAL ASK

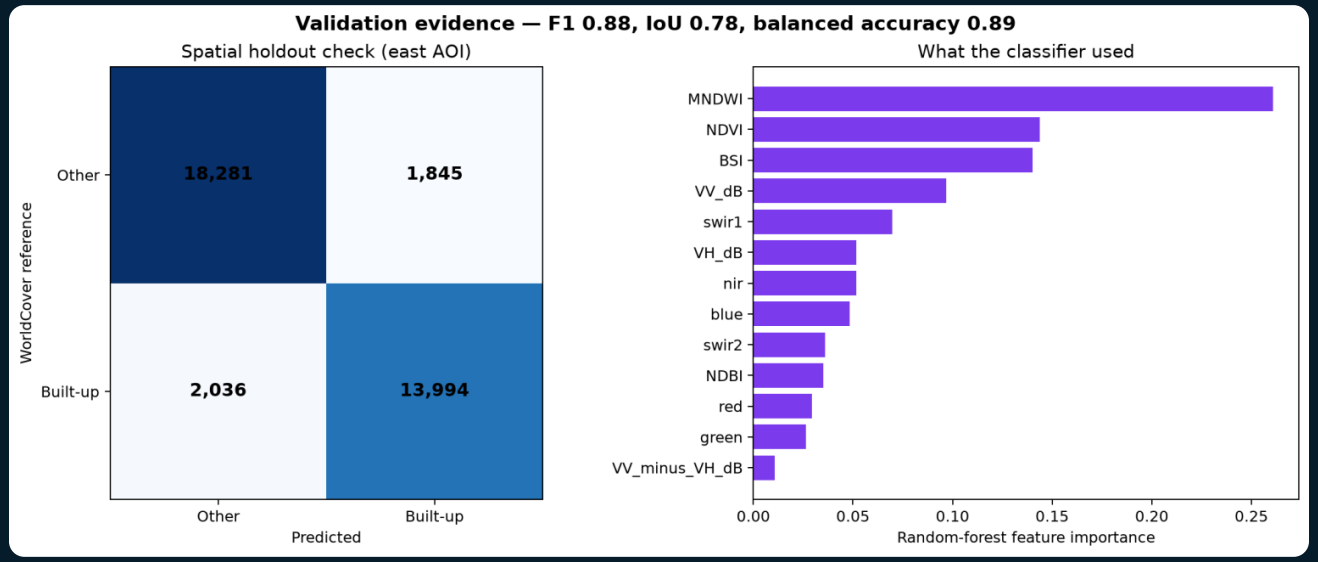
Support one municipal pilot that tests whether HeatReady helps a real team choose where to inspect first — with measurable false alarms, user feedback and updated weights.

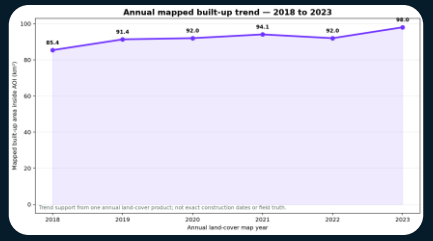
Exact data roles and dates

Source	Dates used	Role	Access / licence
Sentinel-2 L2A	11 May 2018; 10 May 2021; 9 May 2025	Optical features, masks, vegetation	Copernicus open access
Sentinel-1 RTC	10 May 2018; 6 May 2021; 9 May 2025	Radar structure	Microsoft PC; CC BY 4.0
IO/Esri annual LULC	2018–2023	Trend + candidate endpoint change	CC BY 4.0
Landsat 8 C2 L2	10 May 2025	Land-surface temperature	US public domain
ESA WorldCover	2021	Spatial holdout reference	CC BY 4.0
WorldPop Global2	2025	Modelled population	CC BY 4.0
Planet Tanager	15 May 2025	Spectral context + sensitivity test	Open urban archive; CC BY 4.0

Exact item IDs, URLs, processing and attribution are in config.json and docs/DATA_AND_LICENSES.md. The PoC does not use Satellite 813 or restricted imagery.

What the validation can — and cannot — prove





VALIDATION LOGIC

- West strip trains; middle strip tunes; east strip tests against WorldCover.
- The test set is spatially separate and contains more than 10,000 reference pixels.
- The annual time series supports trend interpretation; the 2022 dip shows why it is not construction truth.
- No field survey, air-temperature model, health outcome or municipal acceptance test is claimed.

Operational validation must measure useful inspections and false alarms with a real user.

Fast answers judges may ask for

Why 1 km cells?

They match the population layer and support early area screening; they do not select an individual building.

Why these weights?

They are transparent PoC assumptions: 40% heat, 30% population, 20% growth, 10% lack of vegetation. A real user must review them.

Why not use NDBI alone?

Bright desert can look built-like. Tanager and the optical-radar model show why multiple evidence sources are safer.

Can judges reproduce it?

Yes. The public repository has pinned requirements, a run-all sample notebook, example input/output and the full source pipeline.

What happens if the score is wrong?

No automatic action follows. The cost is wasted inspection time, so the pilot measures false alarms before operational use.

What is next?

User interviews, top-five field checks, local constraints, updated weights and a repeatable test in another Riyadh area.